**Python Basic questions for viva**

**Q1. What is a list?**

A list is an ordered, changeable collection that allows duplicate values. It is written using square brackets [].

**Q2. What is a tuple?**

A tuple is an ordered collection that cannot be changed after creation. It is usually written using parentheses ().

**Q3. What is a dictionary?**

A dictionary stores data as key-value pairs. It is written using curly braces {}.

**Q4. What is a set?**

A set is an unordered collection of unique elements.

**Q5. What is a DataFrame?**

A DataFrame is a two-dimensional, tabular data structure provided by the Pandas library. It contains rows and columns, similar to an Excel spreadsheet or a database table.

**Q6. What is the difference between a list and a DataFrame?**

- A list stores individual items or nested collections.
- A DataFrame stores tabular data with labeled rows and columns.
- Lists are built into Python; DataFrames are provided by Pandas.

**Def. Artificial Intelligence**

Making machines perform intelligent tasks

**def.Machine Learning**

Learning patterns from data

**Slip 1 — Monkey Banana Problem, A* Search and Map Coloring**

1. What is Artificial Intelligence (AI)?

AI is a branch of computer science that enables machines to perform tasks requiring human intelligence.

2. What is the Monkey Banana Problem?

It is an AI problem in which a monkey must use available objects, such as a box, to reach bananas.

3. What is Breadth First Search (BFS)?

BFS is a search algorithm that explores nodes level by level.

4. Which data structure is used in BFS? A queue (FIFO).

5. What is A* Search?

A* is an informed search algorithm used to find a low-cost path from a starting node to a goal.

6. What is the formula for A* Search?

□(□) = □(□) + ℎ(□), where □(□) is the cost so far and ℎ(□) is the estimated cost to the goal.

7. What is a heuristic function?

It estimates the cost or distance from the current state to the goal.

8. What is a Constraint Satisfaction Problem (CSP)?

A problem involving variables, domains of values and constraints.

9. What is the Map Coloring Problem?

It assigns colors to regions so that neighboring regions have different colors.

10. What is backtracking?

It tries a possible solution and goes back when that choice violates a constraint.

**Slip 2 — Water Jug Problem, Backtracking and Game Tree**

1. What is the Water Jug Problem?

It is a state-space problem in which water jugs are used to measure a 2L quantity of water in any of 2 jug.

2. How is a state represented in the Water Jug Problem?

As (□, □), where □ and □ represent the quantities of water in the two jugs.

3. What are the main operations in the Water Jug Problem?

Fill, empty and pour water between jugs.

4. Why is BFS suitable for the Water Jug Problem?

It explores possible states level by level and can find a solution with the fewest steps when each operation has equal cost.

5. What is a state space?

The set of all possible states of a problem.

6. What is a game tree?

A tree representing possible moves and outcomes in a game.

7. What are MAX and MIN players?

MAX tries to maximize the score, while MIN tries to minimize it.

8. What is a leaf node?

A node with no children, often representing a final game outcome.

10. What is the difference between BFS and backtracking?

BFS explores states level by level; backtracking explores a choice and reverses it if it fails.

**Slip 3 — BFS Traversal, Alpha-Beta Pruning and Propositional Logic**

1. What is graph traversal?

It is the process of visiting the vertices of a graph in a systematic way.

2. What is Alpha-Beta Pruning?

It is an optimization of the Minimax algorithm that skips branches that cannot affect the final decision.

3. What is Minimax?

Minimax is an algorithm used in two-player games to choose the best move by considering the opponent's best response.

4. What is alpha ($\alpha$)?

Alpha is the best value found so far for the MAX player.

5. What is beta ($\beta$)?

Beta is the best value found so far for the MIN player.

6. When does Alpha-Beta Pruning occur? When $\alpha \geq \beta$.

7. What is propositional logic?

It is a system of logic that uses statements that are either true or false.

8. What are AND, OR and NOT?

AND is true when both inputs are true; OR is true when at least one input is true; NOT reverses the truth value.

9.. What is a truth table?

A table showing the output of a logical expression for every possible combination of inputs.

**Slip 4 — State-Space BFS, Predicate Logic and Gaussian Naive Bayes**

1. What is state-space search? It represents a problem as states and actions that move from one state to another.

2. What is the initial state? The state from which the search begins.

3. What is the goal state? The desired final state of the problem.

4. What is predicate logic? A type of logic that represents objects, their properties and relationships.

5. What is a predicate?

A statement or function that expresses a property or relationship, such as is_human(Ravi).

6. What is a variable in predicate logic?

A symbol representing an object, such as □ in Human(x).

7. What are quantifiers?

Symbols that specify how many objects satisfy a condition. The universal quantifier ∀ means "for all," and the existential quantifier ∃ means "there exists."

8. What is Naive Bayes? A probabilistic classification algorithm based on Bayes' theorem.

9. What is Gaussian Naive Bayes? A Naive Bayes classifier that assumes numerical features follow a Gaussian (normal) distribution within each class.

10. What is accuracy_score()? A function from sklearn.metrics that calculates the proportion of correct predictions.

**Slip 7 — A* Search, Map Coloring and Tic-Tac-Toe**

1. What is the difference between BFS and A*?

BFS explores level by level, whereas A* uses □(□) + $h$(□) to guide the search.

2. What is the OPEN list in A*?

It contains discovered nodes that still need to be explored.

3. What is the CLOSED list?

It contains nodes that have already been explored, depending on the implementation.

4. What is the goal test?

A check to determine whether the current state is the required destination..

5. What are the variables and domains in Map Coloring?

Variables represent regions; domains represent the available colors.

6. How does Minimax help in Tic-Tac-Toe? It evaluates possible moves and chooses a move that gives the computer the best outcome.

8. Can Tic-Tac-Toe end in a draw? Yes. If all cells are filled and neither player has three marks in a row, the game is a draw.

**Slip 14 — BFS, Supervised vs Unsupervised Learning and Ontology**

1. What is supervised learning?

Machine learning in which the training data includes known labels or target values.

2. Give two examples of supervised learning. Decision Tree and Support Vector Machine (SVM).

3. What is unsupervised learning?

Learning patterns from data without provided target labels.

4. What is the main difference between supervised and unsupervised learning? Supervised learning uses labeled data; unsupervised learning generally uses unlabeled data.

5. What is ontology in AI? A formal representation of concepts, categories, properties and relationships in a domain.

9. What is a class and a subclass? A class represents a category, while a subclass represents a more specific category within it.

10. Give an example of an ontology. Animal is a class, Dog is a subclass, and has_color can be a property.

**Slip 15 — DFS State-Space Search, Clustering Metrics and Knowledge Base**

1. What is Depth First Search (DFS)? DFS explores one branch as deeply as possible before backtracking.

2. Which data structure is used in DFS? A stack or recursion.

3. What is clustering? Grouping similar data points together.

4. What is the Elbow Method? A method that plots inertia against the number of clusters □ to help choose a suitable □.

5. What is the Silhouette Coefficient? A measure of how well a point fits its own cluster compared with other clusters. Values range from $-1$ to $1$.

6. What is a knowledge base? A collection of facts and rules representing knowledge.

7. What is the difference between a fact and a rule? A fact is stored information; a rule specifies what conclusion follows from certain conditions.

**Slip 16 — DLS, Backward Chaining and K-Means Clustering**

2. Why is a depth limit used?

To prevent the search from going too deep or continuing indefinitely along a branch.

3. What happens when the depth limit is reached? The search does not expand that node further.

4. What is backward chaining? An inference method that starts with a goal and works backward to find facts or rules that support it.

5. Where is backward chaining used? In rule-based expert systems, diagnosis and logic programming.

6. What is K-Means? An unsupervised learning algorithm that divides data into  clusters.

7. What is a centroid? The center of a cluster, calculated as the mean of its assigned data points.

8. Why is StandardScaler() used? It standardizes numerical features to have approximately zero mean and unit variance.

9. What is the Elbow Method used for? To help select the number of clusters by observing where the reduction in inertia begins to slow.

10. What is the difference between classification and clustering? Classification predicts predefined labels; clustering discovers groups in data.

**Slip 17 — Best First Search, Propositional Logic and Euclidean Distance**

1. Which data structure is commonly used in Best First Search? A priority queue.

2. What is the difference between Best First Search and BFS? BFS explores nodes level by level; Best First Search selects nodes based on their evaluation or heuristic values.

3. What is propositional logic? Logic that represents statements as true or false.

4. What is Euclidean distance? The straight-line distance between two points.

5. What is the formula for Euclidean distance?

$$□ = \sqrt{\sum_{□=1}^{□} (□_□ - □_□)^2}$$

9. Calculate the distance between (*2,3,4*) and (*5,7,6*).

$$□ = \sqrt{(5-2)^2 + (7-3)^2 + (6-4)^2} = \sqrt{29} \approx 5.385$$

Note: The PDF's stated output is 5.099, but the correct distance for these input vectors is approximately 5.385.

10. Where is Euclidean distance used in machine learning? It is used in algorithms such as K-Nearest Neighbors and K-Means clustering.

**Slip 21 — Iterative Deepening Search, Gaussian Naive Bayes and Map Coloring**

1. What is Iterative Deepening Search (IDS)? It repeatedly performs depth-limited searches, increasing the depth limit each time.

2. What is the difference between DLS and IDS? DLS uses one fixed depth limit; IDS increases the limit repeatedly.

3. What is training data? Data used to teach a machine learning model.

4. What is prediction? Using a trained model to estimate the class or output for new input data.

5.. What is a constraint in Map Coloring? Two neighboring regions must not have the same color.

**Slip 22 — Tower of Hanoi, N-Queens and Rule-Based Expert Systems**

1. What is the Tower of Hanoi Problem? A puzzle involving moving disks between three pegs according to specific rules.

2. What are the rules of Tower of Hanoi? Move one disk at a time, move only the top disk, and never place a larger disk on a smaller disk.

3. What is the minimum number of moves for □ disks?

$$2^□ - 1$$

4. What is the N-Queens Problem? Placing □ queens on an □ × □ chessboard so that no two queens attack each other.

5. What conditions must be satisfied in N-Queens? No two queens can share the same row, column or diagonal.

6. Why is backtracking used in N-Queens? It rejects invalid placements and tries alternative positions.

7. What is an expert system? An AI program that uses a knowledge base and inference rules to make decisions.

8. What is an inference engine? The component that applies rules to known facts to derive conclusions.

**Slip 23 — Hill Climbing, Map Coloring and Supervised Learning**

1. What is Hill Climbing? A local search algorithm that repeatedly moves to a better neighboring state.

2. What is an objective function? A mathematical function that measures the value or quality of a solution.

3. What is a local maximum? A point better than its nearby points, but not necessarily the best point overall.

**Slip 25 — AO* Algorithm, Minimax and Forward Chaining**

1. What is the AO Algorithm?* An informed search algorithm used to solve problems represented by AND-OR graphs.

2. What is an AND-OR graph? A graph in which an AND node requires all its selected subproblems to be solved, while an OR node requires one suitable alternative.

3. What is the difference between A*and AO*?

A* typically searches for a path through a graph; AO* searches for a solution subgraph in an AND-OR graph.

4. What is Minimax? A game-search algorithm that selects moves by maximizing one's outcome while assuming the opponent minimizes it..

5. What is the difference between forward and backward chaining? Forward chaining starts with facts and derives conclusions; backward chaining starts with a goal and searches for supporting facts..

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